# DIPHW2

## 前期准备

**当前服务器信息如下**

| 服务器名 | 内网ip | 外网ip |
| --- | --- | --- |
| diphw2-1 | 192.168.1.6 | 10.119.1.54 |
| diphw2-2 | 192.168.1.9 | 10.119.12.237 |
| diphw2-3 | 192.168.1.14 | 10.119.12.230 |

| 服务名 | 访问地址 | 用户名 | 密码 |
| --- | --- | --- | --- |
| neo4j | 10.119.12.237:7474 | neo4j | G2t5ivrlyDjMCjZ |
| mysql | 10.119.1.54:3306 | root | G2t5ivrlyDjMCjZ |
| hive | 10.119.1.54:10000 | root | G2t5ivrlyDjMCjZ |

密码：G2t5ivrlyDjMCjZ

环境变量：vim /etc/profile

软件及其版本：

| 软件 | 软件版本 |
| --- | --- |
| Java | jdk1.8 |
| jdk21 | 21.0.2 |
| Python3 | 3.6.9 |
| zookeeper | 3.8.4 |
| apache-storm | 2.6.1 |
| kafka | 3.7.0 |
| hive | 4.0.0 |
| neo4j | 5.18.1 |
| mysql | 8.0.33 |
| hadoop | 3.3.6 |
| scala | 2.13.0 |
| spark | 3.5.1 |

**启动指令**

```
# zookeeper
zkServer.sh start

# storm
cd ${STORM_HOME}/bin
run.sh

# hadoop
start-all.sh

# hive
hive --service metastore
hive --service hiveserver2

# spark
/usr/app/spark-3.5.1-bin-hadoop3/sbin/start-all.sh

# mysql
# neo4j
开机自启
```

**服务器hosts设计**

主要是需要指定几个server的名称，不过这些服务器需要在/etc/hosts中先做好映射

注：需要改成自己的内网ip；此外还需要开服务所用到的端口的安全组。这里建议直接对着几个内网ip开放所有端口




**服务器环境变量设计**

在/etc/profile中

```
export JAVA_HOME=/usr/java/jdk21
# jdk21 doesn't have jre
export JRE_HOME=${JAVA_HOME}/jre
export CLASSPATH=.:${JAVA_HOME}/lib:${JRE_HOME}/lib
export PATH=${JAVA_HOME}/bin:$PATH
export ZOOKEEPER_HOME=/usr/app/zookeeper-3.8.4
export PATH=$ZOOKEEPER_HOME/bin:$PATH

export STORM_HOME=/usr/app/storm/apache-storm-2.6.1
export PATH=$STORM_HOME/bin:$PATH

export HIVE_HOME=/usr/app/hive
export PATH=$HIVE_HOME/bin:$PATH

export HADOOP_HOME=/usr/app/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export HDFS_NAMENODE_USER=root
export HDFS_DATANODE_USER=root
export HDFS_SECONDARYNAMENODE_USER=root
export YARN_RESOURCEMANAGER_USER=root
export YARN_NODEMANAGER_USER=root

export SCALA_HOME=/usr/app/scala-2.12.18
export PATH=$PATH:$SCALA_HOME/bin

export SPARK_HOME=/usr/app/spark-3.5.1-bin-hadoop3
export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin
```

## 启动指南

**zookeeper**

首先需要启动zookeepr，storm运行在zookeeper之上。

在每台机器上输入：

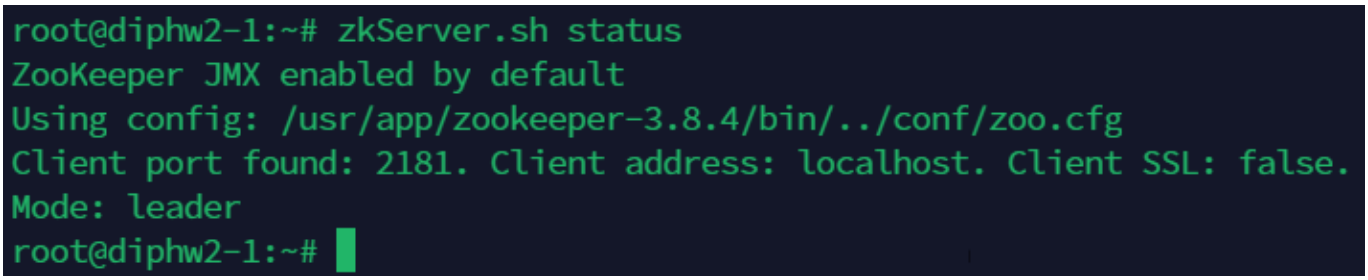
```
zkServer.sh start
```

以启动该机器上的zookeeper。

然后输入：

```
zkServer.sh status
```

查看集群状态。正确启动后的节点应该如下所示，其中Mode为leader或者follower




如果发现异常，建议直接使用：

```
zkServer.sh stop
```

关闭zookeeper，然后在用启动指令重启，直至正常使用。

**Storm**

启动storm前一定要查看zookeeper的状态是否正常。

由于已经编写了启动脚本，所以只需要执行下述指令：

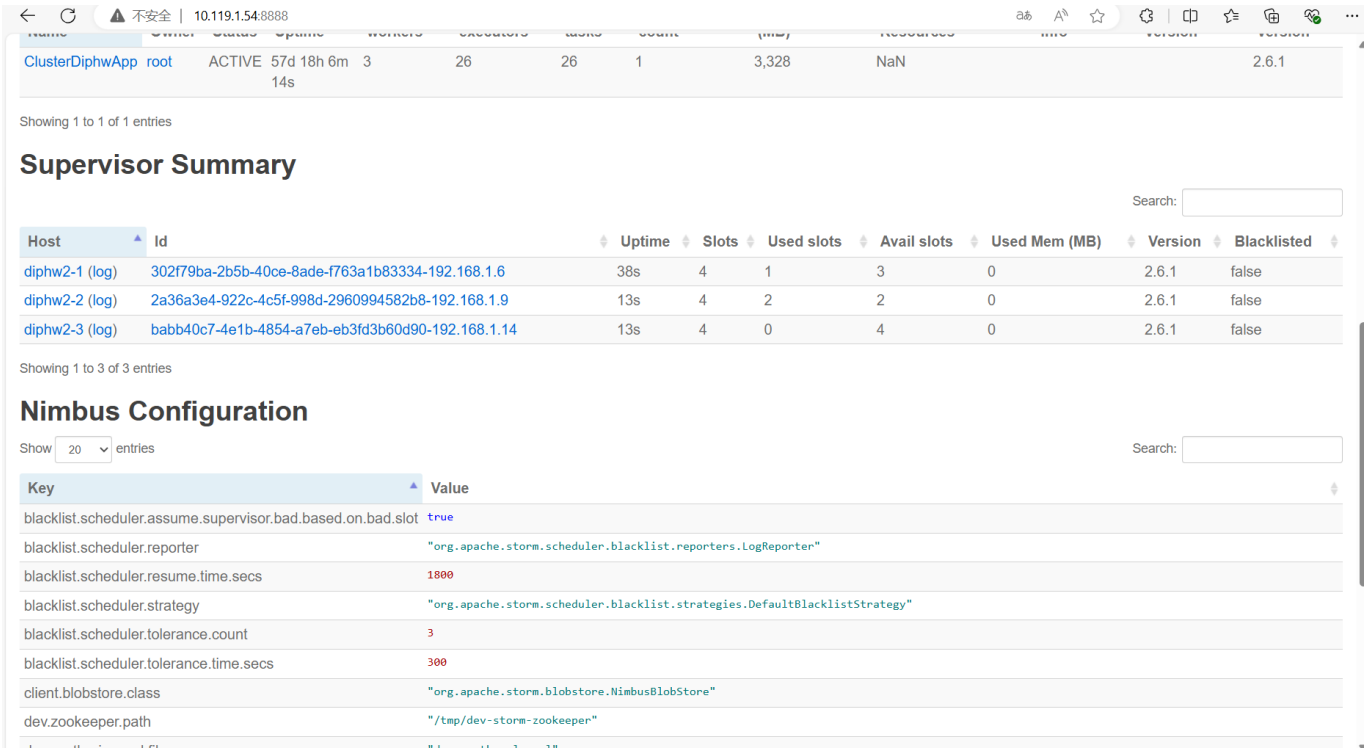
```
cd ${STORM_HOME}/bin
run.sh
```

如果想关闭storm，运行同目录下的

`kill.sh`

注意三台机器都要运行脚本，才能全部启动，不然只能启动一台机器加入集群。

最终可以在主机的8888端口查看storm的运行情况：




### Hadoop

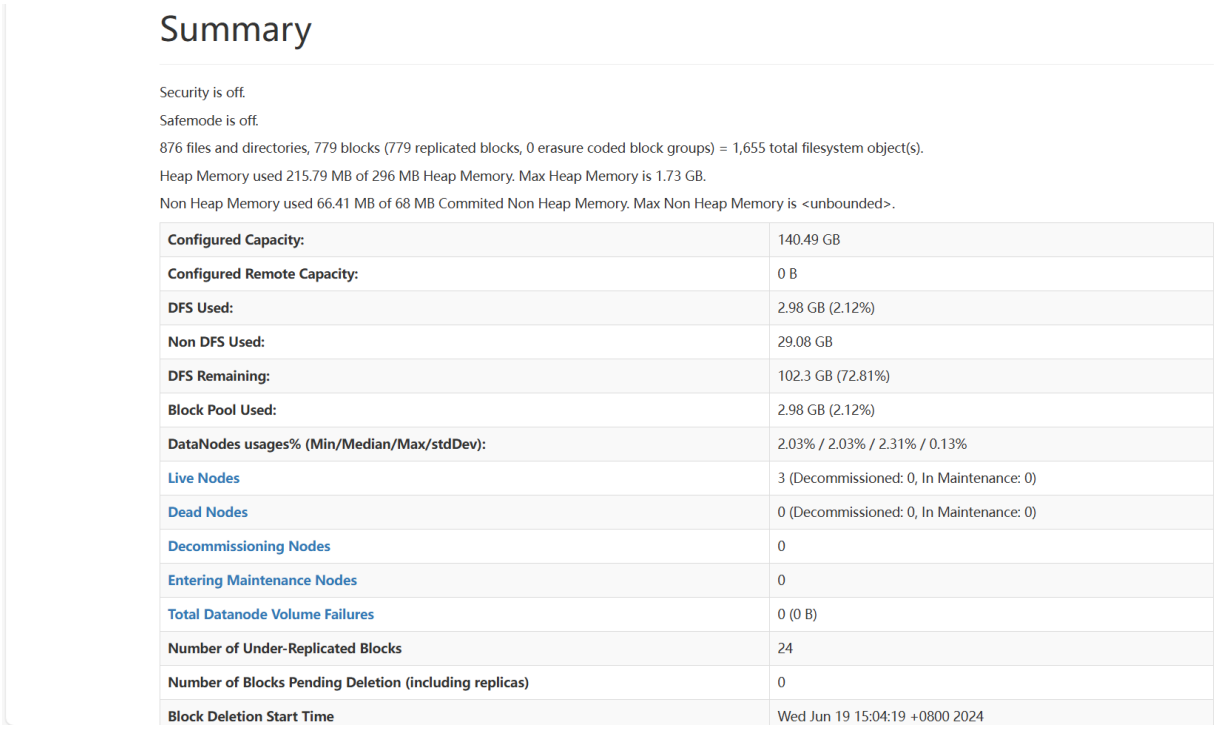
启动Hadoop：

在主机直接输入

```
start-all.sh
```

即可，因为已经配了相关的path。不需要在其他机器上再输入。

启动之后可以在主机的9870端口看到hadoop的相关信息



关闭的话，输入：

```
stop-all.sh
```

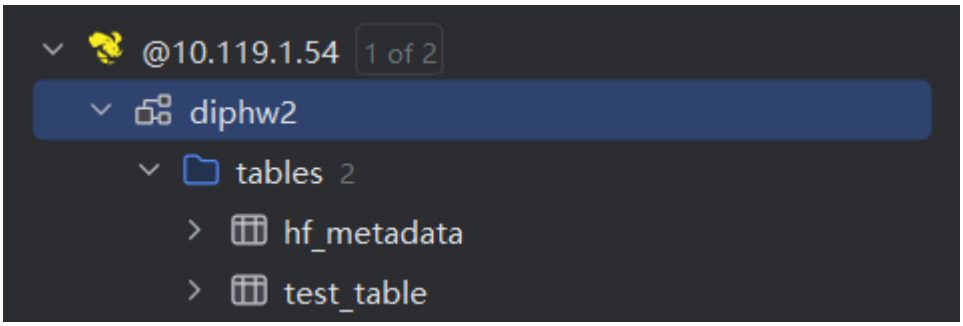
即可。

**Hive**

在主机上依次输入：

```
# 启动数据库
hive --service metastore
# 启动server
hive --service hiveserver2
```

启动后使用工具便可以连接到hive：




**Neo4j**

neo4j开机自启

**Spark**

```
# 启动指令：
/usr/app/spark-3.5.1-bin-hadoop3/sbin/start-all.sh
# 关闭指令
/usr/app/spark-3.5.1-bin-hadoop3/sbin/stop-all.sh
因为被hadoop的环境变量截胡了，所以只能这样了
# shell
spark-shell
# 退出shell
:quit
```

启动后可以在18080端口看到UI。

**Kakfa**

**Kafka with KRaft**

Kafka can be run using KRaft mode using local scripts and downloaded files or the docker image. Follow one of the sections below but not both to start the kafka server.

**Using downloaded files**

Generate a Cluster UUID

```
$ KAFKA_CLUSTER_ID="$(bin/kafka-storage.sh random-uuid)"
```

Format Log Directories

```
$ bin/kafka-storage.sh format -t $KAFKA_CLUSTER_ID -c config/kraft/server.properties
```

Start the Kafka Server

```
$ bin/kafka-server-start.sh config/kraft/server.properties
```

## Storm

一些参考文章：

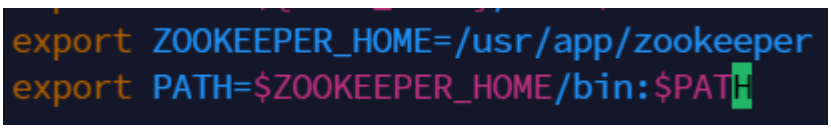
https://blog.csdn.net/u012874209/article/details/52105443

- 官方：https://storm.apache.org/
- 部署参考文档：https://zhuanlan.zhihu.com/p/325661531
- 可以使用java进行开发：https://juejin.cn/post/6844903950034960391?from=search-suggest

**分布式zookeeper安装**

下载对应版本的Zookeeper并解压，在这里解压到$$/usr/app/zookeeper$$下。

配置环境变量至/etc/profile，并运行source /etc/profile




进入zookeeper安装目录下，进入conf目录，复制配置文件 cp zoo_sample.cfg zoo.cfg

```
# The number of milliseconds of each tick
tickTime=2000
# The number of ticks that the initial
# synchronization phase can take
initLimit=10
# The number of ticks that can pass between
# sending a request and getting an acknowledgement
syncLimit=5
# the directory where the snapshot is stored.
# do not use /tmp for storage, /tmp here is just
# example sakes.
dataDir=/usr/local/zookeeper/data
dataLogDir=/usr/local/zookeeper/log
# the port at which the clients will connect
clientPort=2181
# the maximum number of client connections.
# increase this if you need to handle more clients
#maxClientCnxns=60
#
# Be sure to read the maintenance section of the
# administrator guide before turning on autopurge.
#
# http://zookeeper.apache.org/doc/current/zookeeperAdmin.html#sc_maintenance
#
# The number of snapshots to retain in dataDir
#autopurge.snapRetainCount=3
# Purge task interval in hours
# Set to "0" to disable auto purge feature
#autopurge.purgeInterval=1

server.1=diphw2-1:2287:3387
server.2=diphw2-2:2287:3387
server.3=diphw2-3:2287:3387
```

并在每台机器上都指定myid

```
:/usr/local/zookeeper/data# echo "1" >> my_id
```

最后每台机器上运行zhServer.sh start，即可启动集群

指定JAVA路径

在bin/zkENv.sh中，指定java路径



```
if [ "$JAVA_HOME" != "" ]; then
  JAVA="$JAVA_HOME/bin/java"
else
  JAVA=java
fi

JAVA="/usr/java/jdk8/bin/java"
```

**Storm安装**

集群配置：

```
## max.retain.metric.tuples
storm.zookeeper.servers:
     - "diphw2-1"
     - "diphw2-2"
     - "diphw2-3"

nimbus.seeds: ["diphw2-1", "diphw2-2", "diphw2-3"]

# self-added
# Nimbus / Supervisor needs the local disk to store their data(like jar packages,
config files, etc)
storm.local.dir: "/home/storm"

# the port of workers, each worker process uses a port to receive message
supervisor.slots.ports:
        - 6700
        - 6701
        - 6702
        - 6703

ui.port: 8888
```

为了方便启动，编写了启动脚本：

编写脚本

run.sh：

```
cd ${STORM_HOME}/bin
# run nimbus(master)
nohup sh storm nimbus &
# run supervisor(slave)
nohup sh storm supervisor &
# run ui
```

```
nohup sh storm ui &
# run logviewer
nohup sh storm logviewer &
#!/bin/bash

# Kill nimbus process
nimbus_pid=$(ps -ef | grep Nimbus | grep -v grep | awk '{print $2}')
if [ -n "$nimbus_pid" ]; then
    echo "Killing nimbus process (PID: $nimbus_pid)..."
    kill -9 $nimbus_pid
else
    echo "Nimbus process not found"
fi

# Kill supervisor process
supervisor_pid=$(ps -ef | grep Supervisor | grep -v grep | awk '{print $2}')
if [ -n "$supervisor_pid" ]; then
    echo "Killing supervisor process (PID: $supervisor_pid)..."
    kill -9 $supervisor_pid
else
    echo "Supervisor process not found"
fi

# Kill UIServer process
uiserver_pid=$(ps -ef | grep UIServer | grep -v grep | awk '{print $2}')
if [ -n "$uiserver_pid" ]; then
    echo "Killing UIServer process (PID: $uiserver_pid)..."
    kill -9 $uiserver_pid
else
    echo "UIServer process not found"
fi

# Kill LogviewerServer process
logviewer_pid=$(ps -ef | grep LogviewerServer | grep -v grep | awk '{print $2}')
if [ -n "$logviewer_pid" ]; then
    echo "Killing LogviewerServer process (PID: $logviewer_pid)..."
    kill -9 $logviewer_pid
else
    echo "LogviewerServer process not found"
fi
```

赋予脚本可被执行的权力

```
chmod +x /usr/app/storm/apache-storm-2.6.1/bin/run.sh
```

如果storm运行任务时缺少jar包，那么就将对应的jar包放到storm目录的extlib中。

## Hadoop

伪分布式：https://blog.csdn.net/qq_42855570/article/details/115180674

分布式：https://blog.csdn.net/weixin_44259984/article/details/121355692

修改副本数目：https://blog.csdn.net/wsdc0521/article/details/102859900

配置文件路径：/usr/app/hadoop/etc/hadoop/hadoop-env.sh

HDFS：http://10.119.1.54:9870/dfshealth.html#tab-overview

Yarn：http://10.119.1.54:8088/cluster

## Hive

配置文件：/usr/app/hive/conf/hive-site.xml

UI界面：http://10.119.1.54:10002

- 官方：https://hive.apache.org/
- 部署参考文档：https://blog.csdn.net/weixin_46584887/article/details/121481374
- 支持Java、Python等API开发调用
  - Java：https://zhuanlan.zhihu.com/p/469065166
  - Python：https://github.com/dropbox/PyHive

storm集成hive：

https://storm.apache.org/releases/2.1.0/storm-hive.html

**JDBC无法访问hs2**

之前无法通过jdbc连接到hs2，但web UI可以看到hs2，解决方法如下：

首先在hadoop的core-site.xml开启hadoop用户（应该是，没有指定HADOOP_USER的话，就是当前用户）的代理

```
<property>
     <name>hadoop.proxyuser.root.hosts</name>
     <value>*</value>
</property>

<property>
     <name>hadoop.proxyuser.root.groups</name>
     <value>*</value>
</property>
```

在hive的hive-site.xml中，设置用户名和密码

不设置的话，可能会有个默认的？不是很确定

```
    <!-- 指定hiveserver2连接的端口号 -->
    <property>
        <name>hive.server2.thrift.port</name>
        <value>10000</value>
    </property>

    <property>
        <!-- hiveserver2用户名 -->
        <name>beeline.hs2.connection.user</name>
        <value>root</value>
    </property>

    <property>
        <!-- hiveserver2密码 -->
        <name>beeline.hs2.connection.password</name>
        <value>G2t5ivrlyDjMCjZ</value>
    </property>
```

最后连接时，请使用hive-site中使用的url

```
  <!-- 指定hiveserver2连接的host -->
  <property>
      <name>hive.server2.thrift.bind.host</name>
      <value>diphw2-1</value>
  </property>
```

也就是说，不是beeline -u jdbc:hive2://localhost:10000 --verbose=true

localhost可能不识别

```
beeline> !connect jdbc:hive2://localhost:10000 --verbose=true
Connecting to jdbc:hive2://localhost:10000
Enter username for jdbc:hive2://localhost:10000: root
Enter password for jdbc:hive2://localhost:10000: ***************
24/04/08 10:33:38 [main]: WARN jdbc.HiveConnection: Failed to connect to localhost:10000
Could not open connection to the HS2 server. Please check the server URI and if the URI is correct, then ask the administrator to ch
eck the server status. Enable verbose error messages (--verbose=true) for more information.
Error: Could not open client transport with JDBC Uri: jdbc:hive2://localhost:10000: java.net.ConnectException: Connection refused (C
onnection refused) (state=08S01,code=0)
```

```
beeline> !connect jdbc:hive2://diphw2-1:10000 --verbose=true
Connecting to jdbc:hive2://diphw2-1:10000
Enter username for jdbc:hive2://diphw2-1:10000: root
Enter password for jdbc:hive2://diphw2-1:10000: ***************
Connected to: Apache Hive (version 4.0.0)
Driver: Hive JDBC (version 4.0.0)
Transaction isolation: TRANSACTION_REPEATABLE_READ
0: jdbc:hive2://diphw2-1:10000>
```

**Hive在HDFS中的权限问题**

https://blog.csdn.net/yds599089646/article/details/112297124

**Hive插入失败**

https://www.cnblogs.com/cxygg/p/18087973

## Neo4j

官方：https://neo4j.com/docs/

• 支持Java、Python等API开发调用

◦ Java：https://blog.csdn.net/endlessseaofcrow/article/details/80015930

◦ Python：http://py2neo.org/

完全按照下面的教程即可：https://www.techrepublic.com/article/how-to-install-neo4j-ubuntu-server/

需要注意的点：

1. 切记需要修改配置文件，否则外部无法访问，只能本地localhost来访问
2. 第一次默认的用户名和密码：
    1. https://jueee.github.io/2020/08/2020-08-10-Neo4j%E5%88%9D%E5%A7%8B%E5%8C%96%E5%AF%86%E7%A0%81/
    2. neo4j,neo4j
3. 修改后密码：G2t5ivrlyDjMCjZ

## Spark

官方：https://spark.apache.org/

• 支持Java、Python等开发调用，详细参考：https://spark.apache.org/docs/latest/文档的API部 分

安装教程：https://zhuanlan.zhihu.com/p/534723050

本主机安装的是standalone模式的。

**scala安装**

去scala官网下载.tgz文件，解压后，修改脚本中的JAVA_HOME地址（因为scala需要jdk8），一共有两处

```
fi

if [[ -n "$cygwin" ]]; then
    if [[ "$OS" = "Windows_NT" ]] && cygpath -m .>/dev/null 2>/dev/null ; then
        format=mixed
    else
        format=windows
    fi
    SCALA_HOME="$(cygpath --$format "$SCALA_HOME")"
    if [[ -n "$JAVA_HOME" ]]; then
        JAVA_HOME="$(cygpath --$format "/usr/java/jdk8")"
    fi
    TOOL_CLASSPATH="$(cygpath --path --$format "$TOOL_CLASSPATH")"
fi

if [[ -n "$cygwin$mingw$msys" ]]; then
```

```
# reset "$@" to the remaining args
set -- "${scala_args[@]}"

if [[ -z "$JAVACMD" && -n "$JAVA_HOME" && -x "$JAVA_HOME/bin/java" ]]; then
    JAVACMD="/usr/java/jdk8/bin/java"
fi

declare -a classpath_args
```

配置好环境变量后，运行scala即可

```
export SCALA_HOME=/usr/app/scala-2.13.0
export PATH=$PATH:$SCALA_HOME/bin
```

**Spark**

去官网上下载对应的spark版本并解压

按教程来即可

最后需要修改ui的端口（18080）

https://blog.csdn.net/u011095110/article/details/84322266

## MySQL

下载地址：https://downloads.mysql.com/archives/community/

最后一个支持18.04的版本：8.0.33

安装教程：https://blog.csdn.net/weixin_44391770/article/details/132713644

root密码：G2t5ivrlyDjMCjZ

## Kafka

用来定时发送数据流。

kafka集成storm：

1. https://storm.apache.org/releases/2.6.1/storm-kafka-client.html
2. https://juejin.cn/post/6844903950043316231

安装、运行kafka：

1. https://kafka.apache.org/quickstart
2. 使用kraft模式，使用第一种方法运行kafka 三条命令都需要输入

kafka-kraft启动炸了的解决方案：

https://blog.csdn.net/DH2442897094/article/details/120362518

**如何查看kafka中的消息**

```
root@diphw2-1:/usr/app/kafka/kafka_2.13-3.7.0/bin# ./kafka-topics.sh --list --
bootstrap-server localhost:9092
__consumer_offsets
hf-metadata
test
root@diphw2-1:/usr/app/kafka/kafka_2.13-3.7.0/bin# ./kafka-console-consumer.sh --
bootstrap-server localhost:9092 --topic test

^CProcessed a total of 0 messages
root@diphw2-1:/usr/app/kafka/kafka_2.13-3.7.0/bin# ./kafka-console-consumer.sh --
bootstrap-server localhost:9092 --topic hf-metadata
^CProcessed a total of 0 messages
```

如果Kafka出现问题，建议直接重装，比较方便。